# Generative model

- Given data to find $p(x)$, to sample from $p(x)$

原本

$X \to p(x) \to X$

希望

difficulty: $p(x)>0$, $\int_{\mathbb{R}^d} p(x)\,dx = 1$

let $p(x) > 0$

Aisetz: $p(x,\theta) = \dfrac{e^{g(x,\theta)}}{z(\theta)} = \int p\,dx = 1$

let $\int p\,dx = 1$

<u>Score matching</u>: a skill of "generative model"

Score function: $\boxed{S(x)} = \nabla \log p(x) : \mathbb{R}^d \to \mathbb{R}^d$

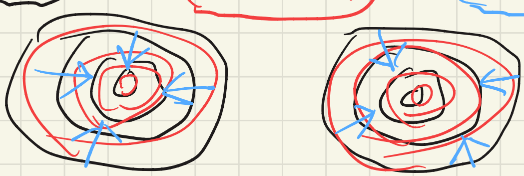
學的東西

(pdf: $p(x) : \mathbb{R}^d \to \mathbb{R}^1$)

pdf

$\log(p(x))$

$S(x) = \nabla \log p(x)$



sample: Gradiant ascent

$x_{n+1} = x_n + \gamma S(c) + \varepsilon$

$(\varepsilon \sim N(0, \sigma^2 I))$

有箭頭就可以
慢慢張出整個sample

# Score matching

- Explicit score matching (ESM)

$$L_{ESM}(\theta) = \mathbb{E}_{x \sim p(x)} \| s(x, \theta) - \nabla \log p(x) \|^2$$

This directly compares the predicted score to the true score, but it's still hard to get $\nabla \log p(x)$

- Implicit score matching (ISM)

$$L_{ISM}(\theta) = \mathbb{E}_{x \sim p(x)} \left[ \| s(x, \theta) \|^2 + 2 \nabla_x \cdot s(x, \theta) \right]$$

Through integration by part,

$L_{ESM} = L_{ISM} + \mathbb{E}_{x \sim p(x)} \| \nabla_x \log(x) \|^2$, where extra term is independent of $\theta$

$$\Rightarrow \quad \theta^* = \underset{\theta}{\operatorname{argmin}} \, L_{ESM} = L_{ISM}$$

Denossing score matching (DSM):

An extension of ESM, DSM is used to learn the score of a pertubed data distribution.

$x_0$ : original data

$p_0(x)$ : original pdf

$x$ : pertubed data

$x = x_0 + \varepsilon$

$p(x)$ = pdf by pertubed data

$$\boxed{L_{DSM} = E_{x_0 \sim p_0(x_0)} E_{(x|x_0) \sim p(x|x_0)} \| S_6(x;\theta) - \nabla_x \log p(x|x_0) \|^2}$$

For gaussian noise distributions, where $x = x_0 + \sigma\varepsilon$

$\varepsilon \sim N(0,1)$, so $\nabla \log p(x|x_0) = -\frac{x - x_0}{\sigma^2} = -\frac{\varepsilon}{\sigma}$

$$\Rightarrow L_{DSM}(\theta) = E \frac{1}{\sigma^2} \| \sigma S_6(x_0 + \sigma\varepsilon ; \theta) + \varepsilon \|^2$$

This allows training with noisy samples without required true data.

## Sliced score matching (SSM, 2019)

By ISM, SSM provides a scalable approximation for high-dimension data by estimating the trace term using Hutchinson's trace estimators.

For a matrix $A$, $tr(A) = E_{v \sim p(v)} [v^T A v]$,

where $v$ is a random vector with $E[vv^T] = I$

$$L_{SSM}(\theta) = E_{x \sim p(x)} \| s(x;\theta) \|^2 + E_{x \sim p(x)} E_{v \sim p(v)} (v^T \nabla v^T s)$$

## Application in Score-Based Generative Model:

this model leverage score matching to train model that generate data by reversing a noise-adding process.

The process involves:

(1) Forward Diffusion Process:

Gradually add noise to the data $x_0$ over time steps $t$, transforming it into near-isotropic

Gaussian noise $x_T \sim N(0, I)$. This is modeled as Markov chain with transitions $q(x_t | x_{t-1})$

(2) Training : A neural network learns the score $S_{\sigma_t}(x_t; \theta) \approx \nabla_{x_t} \log q(x_t)$ for each noise level $\sigma_t$, equivalent to predicting noise $\varepsilon$. DSM or SSM optimizes this efficiently.

(3) Generation : Start from noise $x_T$, the model denoises iteratively using the score function via Langevin dynamics or reverse diffusion.

$$x_{t-1} = x_t + \frac{\sigma_t^2}{2} S_{\sigma_t}(x_i; \theta) + \sigma_t \sqrt{1-\alpha_t}\, z, \quad z \sim N(0,1)$$

The score guides sampling toward high-density region.

(2) Unanswered question

之前看過一個影片是有張長頸鹿的圖片，加一些 noise 之後，基本視覺上看起來相同，但 AI 會認為他是大猩猩，蠻好奇其中的原因。